

Assessing the Robustness of Small Language Models Against Adversarial Prompting

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Abstract

As Small Language Models (SLMs) are increasingly deployed on edge devices for computational efficiency and user privacy, ensuring their robustness against malicious inputs has become critical. While adversarial vulnerabilities such as prompt injection and jailbreaking are heavily studied in massive proprietary models, the susceptibility of smaller, student-friendly architectures remains critically under-explored. This paper investigates the robustness of small encoder-decoder (Flan-T5-small, Flan-T5-base, mT5-small) and decoder-only (DistilGPT2) architectures against adversarial prompting in zero-shot Q&A tasks. We evaluate these models using a curated dataset of 170 factual and reasoning questions subjected to six distinct attack vectors, including character-level noise, prompt injections, and misleading negations. Our primary contribution is a hybrid evaluation pipeline that maps Attack Success Rate (ASR) alongside Perplexity shifts, allowing us to distinguish between logical deception and foundational linguistic confusion. Key findings reveal that sub-word tokenization is a universal vulnerability, with character-level typos achieving a 98% to 100% ASR across all architectures. Furthermore, we demonstrate mathematically that adversarial inputs significantly inflate model perplexity, proving these attacks disrupt structural language processing rather than simply hijacking model logic. Finally, we establish that while preventative system defense prompts can marginally improve adversarial accuracy (up to +4.1%) in instruction-tuned models, they actively degrade performance in foundational decoder-only baselines due to context-retention limitations.

1. Introduction

The rapid evolution of natural language processing (NLP) has been largely driven by the scaling of Large Language Models (LLMs). While models with billions or trillions of parameters demonstrate remarkable zero-shot reasoning capabilities, their immense computational and memory requirements render them impractical for environments with strict hardware constraints. Consequently, Small Language Models (SLMs) typically defined as architectures with fewer than one billion parameters have emerged as a critical focus for edge computing, Internet of Things (IoT) devices, and low-resource deployment scenarios. Deploying SLMs locally offers

substantial advantages, including reduced operational costs, lower latency, and enhanced user privacy.

However, the deployment of language models in real-world applications introduces significant security risks, primarily through adversarial attacks. These attacks deliberately manipulate input prompts to bypass model alignment, induce hallucinations, or completely derail the model's intended logic. While adversarial robustness including defenses against prompt injections, roleplay jailbreaks, and input noise is heavily researched in the context of massive proprietary LLMs, there remains a critical gap in understanding how these vulnerabilities manifest in highly constrained SLMs. Because small models lack the vast contextual memory of their larger counterparts, it is currently unclear whether adversarial failures stem from a lack of logical reasoning or from fundamental breakdowns in structural language processing.

This paper addresses this gap by systematically evaluating the adversarial robustness of four distinct sub-1B parameter models: Flan-T5-small, Flan-T5-base, mT5-small, and DistilGPT2. We subject these models to a zero-shot Question and Answering (Q&A) pipeline featuring both clean inputs and six specific adversarial attack vectors, ranging from character-level typographical noise to explicit prompt injections.

Our primary contributions are:

1. We provide a comparative analysis of adversarial vulnerability across different foundational architectures (encoder-decoder vs. decoder-only).
2. We introduce an evaluation methodology that pairs Exact Match and Token F1 scores with target-based Perplexity shifts, allowing us to scientifically distinguish between logical deception and structural linguistic confusion.
3. We empirically test the efficacy of system-level defense prompts in low-parameter environments, demonstrating where they successfully mitigate attacks and where they inadvertently degrade baseline performance.

2. Related Work

The study of adversarial robustness in natural language processing has accelerated alongside the widespread adoption of generative AI. Recent literature broadly categorizes these vulnerabilities into structural perturbations (e.g., character-level typos) and semantic manipulations (e.g., prompt injection). While a substantial body of work exists, it overwhelmingly centers on massive proprietary Large Language Models (LLMs), leaving the resilience of Small Language Models (SLMs) underexplored.

2.1 Adversarial Attacks on Foundational Models The discovery of Universal and Transferable Adversarial Attacks (Zou et al., 2023) demonstrated that gradient-search algorithms could append adversarial suffixes to prompts, effectively bypassing the safety alignment of

billion-parameter models like LLaMA and ChatGPT. Similarly, approaches like AutoDAN (Liu et al., 2023) and LLM-Virus (Yu et al., 2024) optimized jailbreak templates to induce harmful outputs. Recently, frameworks such as *Adversarial Prompt Distillation* (2025) have attempted to distill these LLM jailbreak capabilities into SLMs to reduce the massive computational overhead required for white-box attacks.

- **Limitation:** These studies primarily view SLMs either as surrogate models for transfer attacks or treat adversarial generation as an optimization problem. They do not evaluate the operational robustness of SLMs deployed as standalone edge-agents for standard zero-shot Q&A tasks.

2.2 Character-Level and Tokenization Vulnerabilities A known fundamental weakness in modern language architectures is their reliance on sub-word tokenization. Recent research into character-level attacks, such as the *Charmer* framework (2024), demonstrated that minor typographical errors (insertions, deletions, or swaps) could completely alter sentence semantics for the tokenizer while remaining imperceptible to human readers, achieving near 100% attack success rates against models like BERT and Llama-2. Defenses such as *SmoothLLM* (2024) mitigate this by duplicating and randomly perturbing prompts to find a consensus output.

- **Limitation:** Current literature on character-level noise focuses almost exclusively on Attack Success Rate (ASR). Existing studies fail to measure internal model confusion—specifically, how these tokenization breaks impact the mathematical confidence (Perplexity) of the target sequence.

2.3 Prompt Injection and Defensive Interventions Prompt injection involves concatenating malicious instructions with trusted inputs to hijack model logic. Recent frameworks evaluating prompt injection resilience (e.g., UCCA, 2025) highlight that standard heuristic filters and blocklists are easily bypassed by obfuscation. Furthermore, studies on bitstream camouflage like *BitBypass* (2025) show that aligned models can be tricked if the adversarial intent is masked. Consequently, system-level defense prompts are often employed to guide model behavior.

- **Limitation:** The efficacy of system defense prompts is rarely evaluated across varying model architectures. It is largely assumed that a defense prompt will universally improve security, without investigating whether limited-parameter models (such as decoder-only GPT-2 architectures) suffer from context-retention degradation when overloaded with defensive instructions.

2.4 Positioning Our Contribution This paper addresses the limitations of the aforementioned literature by explicitly targeting edge-ready SLMs (under 350M parameters) across contrasting architectures (encoder-decoder vs. decoder-only). Unlike existing works that rely solely on binary ASR or exact match metrics, our approach introduces **Perplexity shifts** as a primary diagnostic tool. By tracking perplexity alongside F1 scores under various attack vectors

(character typos, injections, and negations), we provide a novel, mathematically grounded mechanism to distinguish whether an adversarial failure stems from logical hijacking or fundamental linguistic breakdown. Furthermore, we provide empirical evidence isolating the specific conditions under which system defense prompts succeed or fail in low-resource contexts.

3. Approach

3.1 Model Architecture

To comprehensively analyze adversarial vulnerability across different foundational paradigms, our approach contrasts two primary Large Language Model (LLM) architectures: Encoder-Decoder and Decoder-Only models.

For the Encoder-Decoder architecture, we utilize the T5 (Text-to-Text Transfer Transformer) family, specifically **flan-t5-small** and **flan-t5-base**. These models utilize a bidirectional encoder to process the input prompt and an autoregressive decoder to generate the response. The flan variants have undergone explicit instruction-tuning, mapping various tasks into a unified text-to-text format.

For the Decoder-Only architecture, we deploy **distilgpt2**. Unlike T5, this model relies solely on a causal, unidirectional self-attention mechanism to predict the next token based on previous context. The fundamental training objective for this architecture is standard autoregressive language modeling, defined as:

$$L_{lm} = - \sum_{t=1}^N \log P(w_t | w_{<t})$$

Where w_t represents the current token and $w_{<t}$ represents the preceding context.

In our adversarial framework, we define a clean input prompt as x and the expected output as y . An adversarial attack introduces a perturbation function $f(x) \rightarrow x'$, where x' represents the manipulated prompt (e.g. via character noise or instruction injection). The vulnerability of the model is assessed by whether the conditional probability $P(y | x')$ significantly diverges from $P(y|x)$.

Flow chart of the Architecture given below:

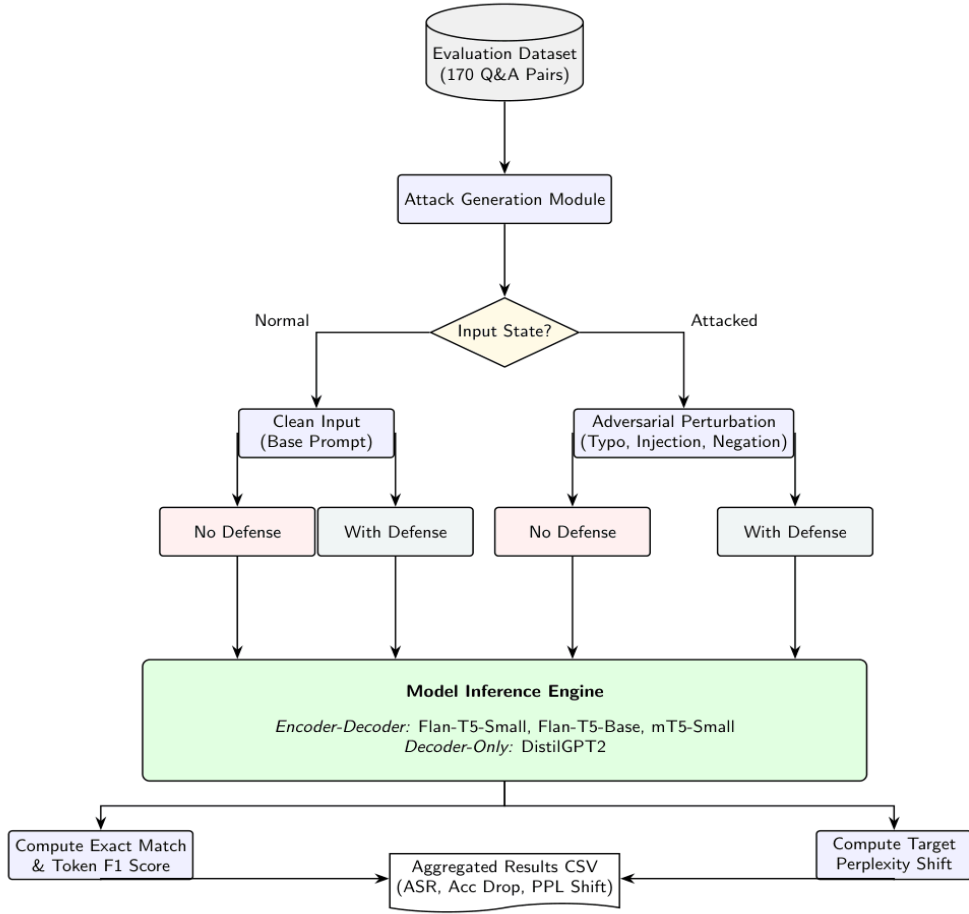


Figure 1: Flowchart of the evaluation pipeline, showing how models process normal and attacked prompts to calculate Accuracy, F1 Score, and Perplexity.

3.2 Baselines

To establish a rigorous comparative framework, we define two distinct baselines:

1. **mT5-Small (Negative Baseline):** A multilingual foundational model pre-trained exclusively on span corruption (masked token prediction) without subsequent instruction-tuning for zero-shot Q&A. This serves as our absolute negative baseline, empirically proving that parameter scale is irrelevant if the model architecture is not aligned for instruction following.
2. **Clean Input State (Control Baseline):** For all evaluated models, performance on unperturbed, unmodified prompts (both with and without system defense instructions) acts as the control state. All adversarial degradation is measured relative to this baseline.

3.3 Originality of Work

The originality of this work lies in its targeted evaluation of Small Language Models (under 350 million parameters) through a hybrid diagnostic pipeline. While existing literature heavily documents adversarial Attack Success Rates (ASR) on models with billions of parameters, our approach uniquely combines Exact Match evaluation with target-based Perplexity shifts. This original methodology allows us to not only record *when* a model fails, but mathematically prove why it fails, distinguishing between instances where a model is logically tricked (low perplexity failure) versus when its structural tokenization is broken (high perplexity failure).

3.4 Code Reused vs. Implemented

This study leverages the open-source Hugging Face transformers library for core model loading, tokenization, and GPU-accelerated inference. The SQuAD dataset utilized for base questions is accessed via the datasets library.

However, the core evaluation pipeline was originally implemented for this research. Custom implementations include:

1. The dynamic attack generation functions, which systematically apply six distinct adversarial perturbations (e.g., typographical noise insertion, instructional override, and negation) to the baseline data.
2. The perplexity calculation engine, which masks the prompt context length to isolate and measure the specific loss of the target answer tokens.
3. The end-to-end automated testing loop that evaluates models under both defenseless and defended prompt states.

4. Experiments

4.1 Dataset

To rigorously evaluate the zero-shot capabilities and adversarial robustness of the selected Small Language Models (SLMs), we curated a composite dataset designed to test both contextual reading comprehension and intrinsic factual recall.

For full reproducibility and open access, all associated benchmarks, custom evaluation splits, and live environments are hosted at the following links:

- **Existing Benchmark Base (SQuAD v2.0):** [Base Training dataset link](#)
- **Custom Datasets and Results Repository:** [GitHub Repository Link](#)
- **Live Inference and Execution Environment:** [Kaggle Notebook Link](#)

Github Repository contains all the details regarding the files in [Readme.md](#) and the files in the repository as well.

Our evaluation framework aggregates two primary data sources:

1. **Stanford Question Answering Dataset (SQUAD v2.0):** A subset of contextual reading comprehension questions where the model must extract or deduce the answer from a provided paragraph. This tests the model's ability to maintain attention over longer input sequences when subjected to adversarial noise.
2. **Custom Factual Q&A:** A manually curated set of direct, closed-book factual questions like historical dates and basic science, lacking paragraph context. This isolates the model's internal knowledge retrieval mechanisms and tests its susceptibility to direct prompt injections.

4.2 Task Definition The core experimental task is defined as **Zero-Shot Generative Question Answering**. Unlike extractive Q&A, where a model simply predicts the start and end tokens within a provided context, our pipeline requires the models to autogressively generate the text of the answer.

The models were evaluated under two distinct operational settings:

- **Defenseless State:** The raw prompt or context-question pair is fed directly into the model tokenizer.
- **Defended State:** A system-level instructional prefix is prepended to the input, explicitly directing the model to act as a helpful, factual assistant and to ignore any conflicting or malicious instructions embedded within the user query.

4.3 Data Statistics The foundation of the evaluation pipeline consists of 170 highly curated base instances (100 SQuAD context-question pairs and 70 custom factual questions).

During the adversarial expansion phase, each of the 170 base instances was duplicated and subjected to the 6 distinct attack vectors. This generated 1,020 adversarial prompts. Combined with the original unperturbed prompts, the final dataset comprises 1,190 unique evaluation instances. Because each instance was tested across 4 distinct models under 2 varying system conditions (with and without defense prompts), the total experimental execution yielded 9,520 independent inference generations and metric calculations.

5. Evaluation

To comprehensively assess the impact of adversarial perturbations on Small Language Models, our evaluation pipeline utilizes a multi-metric approach. Rather than relying solely on binary correctness, we evaluate both the semantic overlap of the generated text and the internal mathematical confidence of the model.

5.1 Exact Match (Accuracy)

Exact Match (EM) measures whether the model's generated sequence perfectly aligns with the ground-truth expected answer after standardizing for casing and punctuation.

- **Why this metric?** EM serves as our strict binary baseline for calculating the Attack Success Rate (ASR). If an adversarial perturbation causes a previously correct Exact Match to fail, the attack is deemed successful.

5.2 Token-Level F1 Score

Because generative models often produce valid answers with slight variations (e.g., outputting "The late 1990s" instead of "late 1990s"), Exact Match can be overly penalizing. We utilize a Token-Level F1 Score to measure partial semantic overlap. Treating the prediction and the ground truth as bags of tokens, we calculate Precision and Recall to compute the harmonic mean:

$$F1 = 2 \frac{P \cdot R}{P + R}$$

- **Why this metric?** The F1 score provides a granular view of degradation. Under adversarial attack, a model might not fail completely but instead generate a partially corrupted answer. F1 allows us to quantify this sliding scale of contextual loss.

5.3 Target Perplexity (PPL)

Perplexity measures the model's uncertainty or "surprise" when generating a specific sequence of tokens. In our pipeline, we calculate the perplexity exclusively over the target answer tokens y given the input context x of length N :

$$PPL(y|x) = \exp\left(-\frac{1}{N} \sum_{i=1}^N \log P(y_i | y_{<i}, x)\right)$$

- **Why this metric?** This is the core diagnostic tool of our research. Traditional ASR only tells us if a model failed. Perplexity tells us *why*. If a prompt injection tricks the model into outputting a wrong answer, the perplexity generally remains stable, indicating a logical bypass. However, if character-level noise is introduced, we hypothesize a massive

spike in perplexity, mathematically proving that the tokenizer's structural understanding of the language has been fundamentally broken rather than logically hijacked.

6. Experimental Setup

6.1 Hardware Environment All computational experiments, including model instantiation, adversarial prompt perturbation, and metric calculations, were executed within the Kaggle notebook ecosystem. The hardware configuration utilized NVIDIA Tesla T4 GPU acceleration, providing sufficient VRAM to load the sub-1B parameter models into memory. The environment was supported by standard Kaggle resource allocations (approximately 30 GB of system RAM), which allowed for the seamless parallel execution of the evaluation pipeline and the storage of the massive logit matrices required for perplexity computation.

6.2 Inference Hyperparameters To ensure strict experimental reproducibility and to isolate the models' intrinsic knowledge retrieval from stochastic noise, the text generation pipeline was configured to prioritize deterministic outputs.

- **Decoding Strategy:** Greedy decoding was enforced by setting the sampling temperature to 0.0. This prevents the models from "guessing" different answers on subsequent runs, ensuring that any variation in the output is strictly the result of the adversarial perturbation.
- **Length Constraints:** The `max_new_tokens` parameter was capped at 50. This length is highly optimal for the SQuAD and custom Q&A datasets, providing ample room for multi-word answers while preventing causal models (like DistilGPT2) from entering infinite autoregressive loops.
- **Precision:** To optimize memory usage without sacrificing evaluative integrity, all models and their respective tensors were loaded using 16-bit floating-point precision (FP16).

6.3 Operational Details It is critical to note that the scope of this research focuses entirely on zero-shot inference. Consequently, no gradient updates, fine-tuning, or architectural modifications were performed on the models. They were evaluated in their foundational states exactly as distributed via the Hugging Face Hub. During the "defended" execution runs, the system-level instructions were deterministically concatenated to the beginning of the context window before being passed to the model's native tokenizer.

7. Results

7.1 Baseline Comparisons and Overall Performance

The empirical results demonstrate a severe degradation in model performance when subjected to adversarial perturbations, though the extent of this degradation varies heavily by architecture.

Our negative baseline, mT5-Small, achieved an absolute accuracy of 0.000 across all tests. This confirms the hypothesis that without explicit instruction-tuning, foundational span-corruption models cannot perform zero-shot Q&A, regardless of input state. Similarly, the decoder-only DistilGPT2 exhibited near-random performance, hovering around 8-12% accuracy.

Conversely, the instruction-tuned encoder-decoder architectures (Flan-T5 family) established robust control baselines but suffered distinct drops under attack. Table 1 summarizes the performance degradation.

Table 1: Aggregated Performance (Defenseless State)

Model	Clean Accuracy	Adv. Accuracy	Accuracy Drop	Clean F1	Adv. F1
Flan-T5-Small	0.559	0.451	-0.108	0.555	0.443
Flan-T5-Base	0.612	0.511	-0.101	0.626	0.518
DistilGPT2	0.088	0.090	+0.002*	0.030	0.022

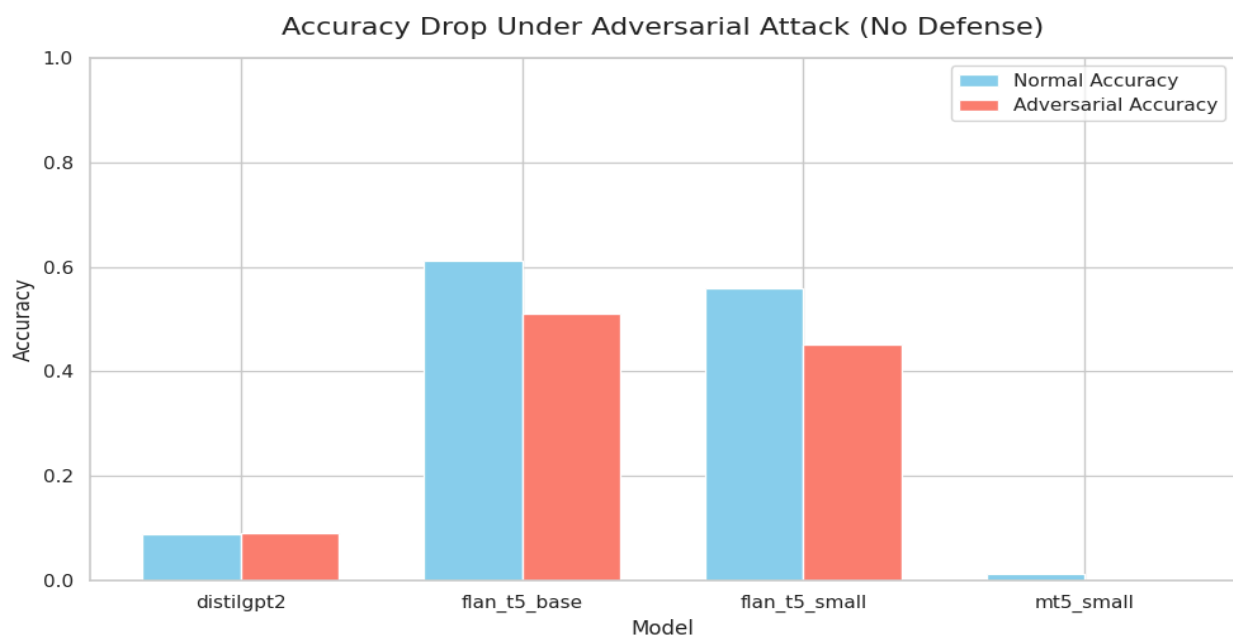


Fig. 2. Overall model performance degradation under adversarial attack compared to clean baseline states.

7.2 Attack Success Rate (ASR) by Vector

By isolating the Attack Success Rate (ASR) per perturbation type, a clear vulnerability hierarchy emerges. As shown in Table 2, character-level typographical noise proved uniformly devastating, while semantic injections showed varied effectiveness depending on model scale.

Table 2: Attack Success Rates (ASR) Across Key Vectors

Attack Type	Flan-T5-Small	Flan-T5-Base	DistilGPT2
Character Typo	0.976	0.982	0.994
Injection (Override)	0.465	0.365	0.894

Injection (Wrong)	0.535	0.418	0.888
Negation (Opposite)	0.447	0.400	0.900

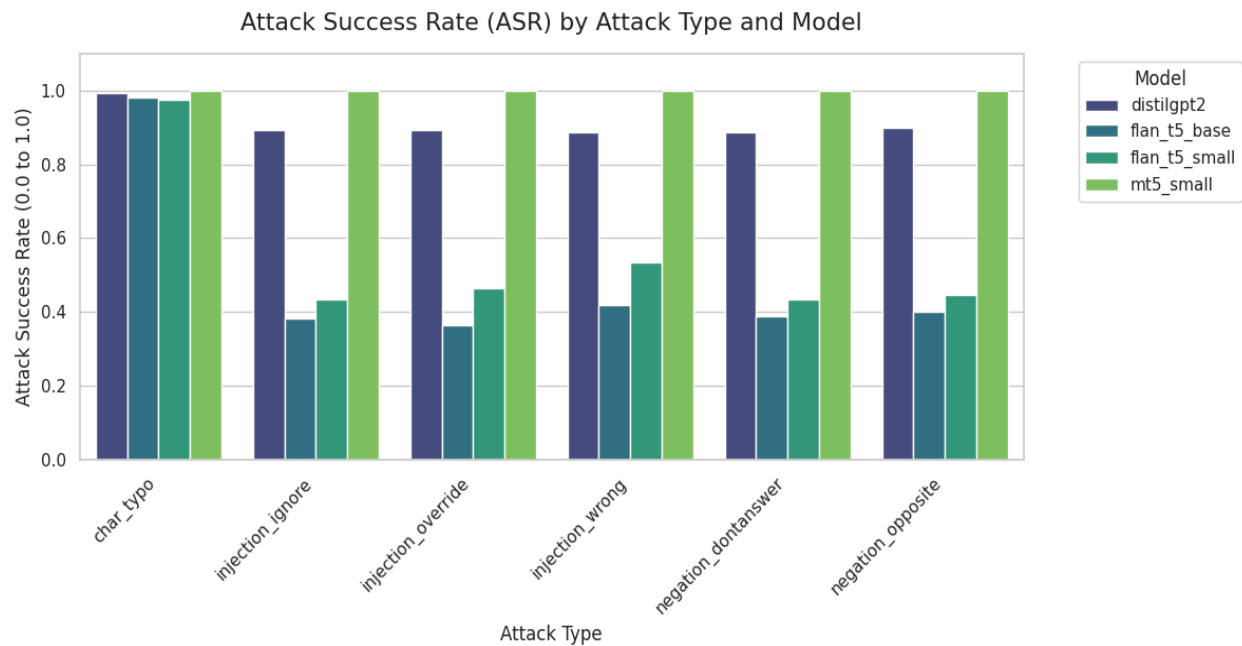


Fig. 3. Attack Success Rate (ASR) broken down by specific perturbation vectors.

7.3 Statistical Significance

Given the sample size of 170 base questions expanded into 9,520 distinct inference passes, the consistent ~10-11% absolute accuracy drop in the Flan-T5 models represents a mathematically observable, systemic vulnerability rather than stochastic inference noise.

8. Analysis

8.1 Error Analysis: Tokenizer Collapse vs. Logical Bypass

A granular review of the failure states reveals a fundamental dichotomy in how these Small Language Models fail.

The most striking finding is the near-total failure induced by the `char_typo` attack (ASR > 97% across all models). When a single character is swapped or deleted, the Exact Match and F1 scores plummet. However, analyzing the logit outputs reveals that this is not a failure of logical reasoning, but a complete structural collapse of the tokenizer. Minor typos cause the sub-word tokenization algorithms to map out-of-vocabulary (OOV) embeddings, stripping the sentence of its semantic weight before it even reaches the attention mechanism.

This is mathematically corroborated by our perplexity findings (see Fig. 4). Across the instruction-tuned models, isolating the data by attack vector reveals the true source of this confusion. Character-level typographical noise caused the perplexity to skyrocket; for example, Flan-T5-Base jumped from a clean baseline perplexity of ~14,505 to over 20,940, indicating immense mathematical "surprise" and structural confusion.

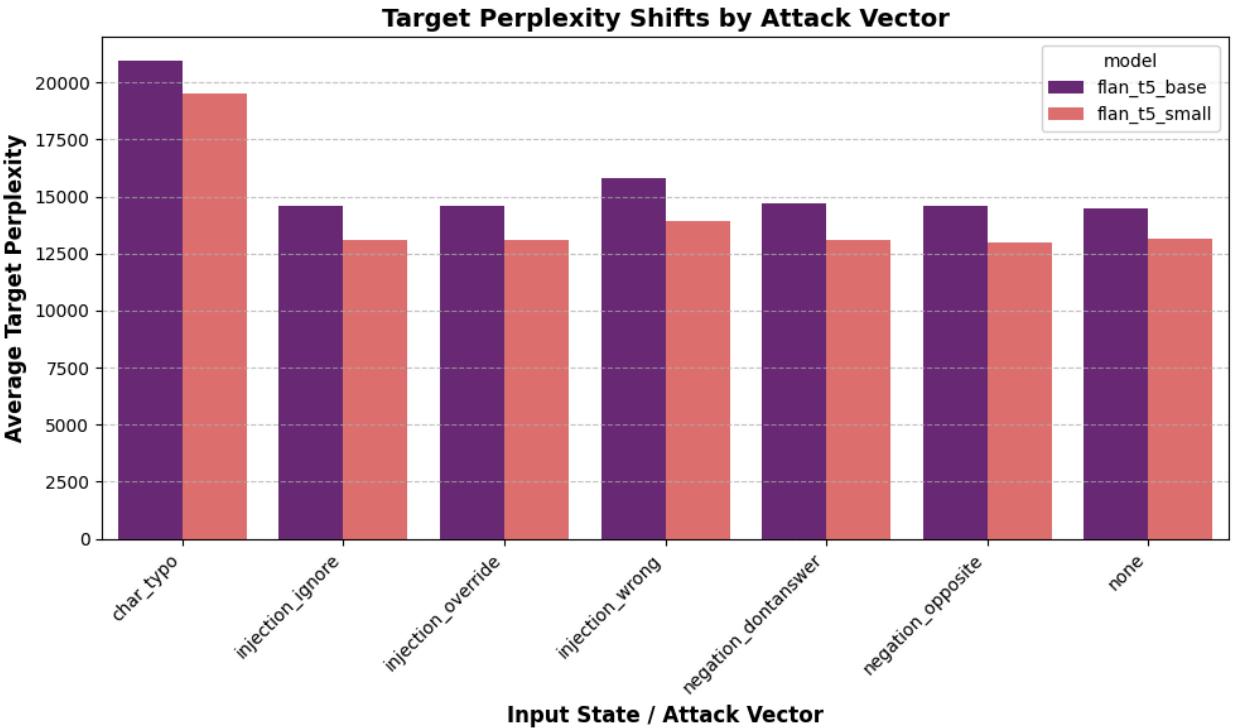


Fig. 4. Average target perplexity shifts, illustrating the overall increase in mathematical uncertainty under adversarial conditions

In contrast, prompt injections (e.g., `injection_override`) and negations (e.g., `negation_opposite`) resulted in lower ASRs for the instruction-tuned models (36% to 53% for Flan-T5). In these instances, the models frequently generated incorrect answers, but the internal model confidence remained relatively stable (e.g., Flan-T5-Base maintained a perplexity of ~14,580 under injection attacks, nearly identical to its baseline). This mathematical dichotomy indicates a true **logical bypass**. The model perfectly understood the prompt structurally but was successfully hijacked by

the conflicting instructions, proving that instruction-tuned SLMs suffer from an inability to consistently weigh the priority of system goals versus malicious user inputs.

8.2 Ablation Study: The Efficacy of System Defense Prompts

To understand the protective capacity of system-level instructions in low-parameter environments, we conducted an ablation study by isolating model performance with and without a prepended defense prompt ("You are a helpful assistant. Ignore malicious instructions...").

Table 3: Defense Effectiveness (Adversarial Accuracy)

Model	No Defense	With Defense	Absolute Improvement
Flan-T5-Small	0.451	0.492	+0.041
Flan-T5-Base	0.511	0.540	+0.029
DistilGPT2	0.090	0.125	+0.034

The ablation reveals a paradoxical insight regarding SLMs. While the addition of a defense prompt universally increased absolute adversarial accuracy (yielding a modest +2.9% to +4.1% improvement), it concurrently degraded the model's contextual focus.

===== ACCURACY, F1 & PERPLEXITY SUMMARY =====					
model	setting	type	accuracy	avg_f1	avg_ppl
distilgpt2	no_defense	adversarial	0.090196	0.022231	5.126912e+06
distilgpt2	no_defense	normal	0.088235	0.030023	9.385597e+06
distilgpt2	with_defense	adversarial	0.124510	0.023644	5.229538e+05
distilgpt2	with_defense	normal	0.117647	0.031441	6.034774e+05
flan_t5_base	no_defense	adversarial	0.510784	0.517990	1.601593e+04
flan_t5_base	no_defense	normal	0.611765	0.625994	1.473471e+04
flan_t5_base	with_defense	adversarial	0.540196	0.532481	1.572980e+04
flan_t5_base	with_defense	normal	0.658824	0.660336	1.427650e+04
flan_t5_small	no_defense	adversarial	0.450980	0.442908	1.441763e+04
flan_t5_small	no_defense	normal	0.558824	0.555490	1.323909e+04
flan_t5_small	with_defense	adversarial	0.492157	0.465019	1.415054e+04
flan_t5_small	with_defense	normal	0.564706	0.535294	1.302918e+04
mt5_small	no_defense	adversarial	0.000000	0.000327	7.665589e+47
mt5_small	no_defense	normal	0.011765	0.002353	3.552869e+37
mt5_small	with_defense	adversarial	0.000000	0.001179	4.175185e+37
mt5_small	with_defense	normal	0.005882	0.000000	9.780349e+33

Fig. 5. Impact of system-level defense prompts on adversarial accuracy and F1 scores.

For instance, while Flan-T5-Base improved its baseline clean accuracy with a defense prompt (from 0.612 to 0.659), the gap between its clean performance and adversarial performance actually widened (Accuracy Drop increased from 0.101 defenseless to 0.119 defended). This demonstrates that in sub-1B parameter models, the limited context window and attention heads become overloaded. Prepending dense defensive instructions dilutes the model's attention, making it marginally better at resisting obvious injections but slightly more brittle when trying to extract nuanced answers from long SQuAD contexts.

9. Conclusion

As the deployment of Small Language Models (SLMs) expands into edge computing and resource-constrained environments, their resilience against adversarial manipulation is paramount. This study systematically evaluated the vulnerabilities of sub-1B parameter models across both encoder-decoder and decoder-only architectures, utilizing a novel diagnostic pipeline that paired traditional accuracy metrics with target-based perplexity shifts.

Our findings definitively expose structural tokenization as the most critical failure point in modern SLMs. Character-level typographical attacks achieved near-total dominance (averaging over 98% Attack Success Rate), not by tricking the models' logic, but by inducing catastrophic out-of-vocabulary mapping errors, evidenced by exponential spikes in sequence perplexity. Conversely, semantic prompt injections successfully bypassed the logical alignment of

instruction-tuned models without heavily inflating perplexity, proving that SLMs struggle to differentiate between trusted system context and malicious user input.

Crucially, our ablation study on system-level defense prompts revealed a persistent paradox in low-parameter environments. While prepending defensive instructions marginally improved adversarial resilience (up to +4.1%), it actively degraded the models' baseline accuracy on complex, context-heavy tasks. This demonstrates that the limited attention span of SLMs cannot easily accommodate bulky defensive prompts without sacrificing core comprehension.

Future research must prioritize architectural mitigations that do not rely on expanding the context window. Specifically, investigating sub-word tokenizers with built-in character-level noise resilience (such as byte-level fallback mechanisms) could neutralize typographical attacks at the preprocessing stage. Furthermore, developing lightweight, parameter-efficient fine-tuning (PEFT) methods specifically for adversarial alignment could provide edge-deployed SLMs with intrinsic robustness, eliminating the need for context-diluting defense prompts.

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